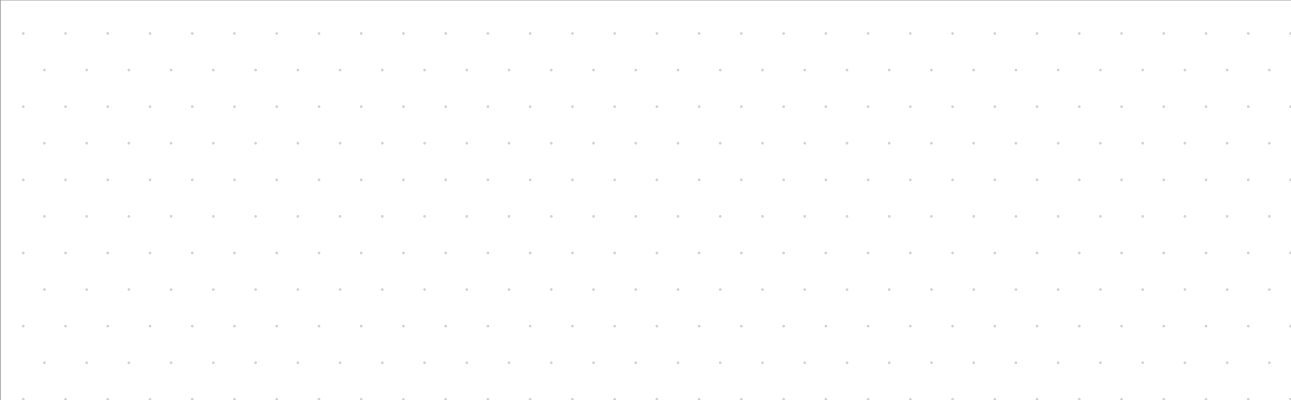


Use small green triangles, small blue rhombi, or both. Pieces must not overlap.

Problem 1: Use exactly 6 green triangles and 2 blue rhombi. Join all eight blocks into one shape along whole sides. Make a shape with the shortest boundary you can, and one with the longest. Count the uncovered block sides, including any around a hole, and sketch both shapes.

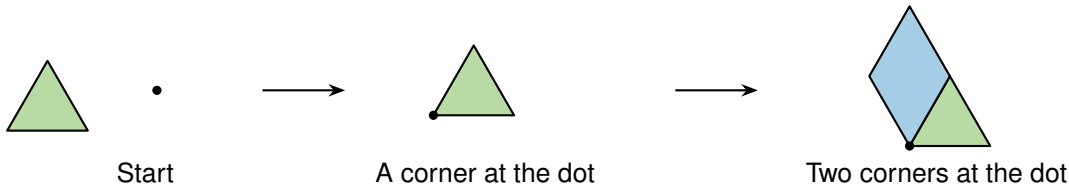


Boundary: _____ block sides



Boundary: _____ block sides

Problem 2: On the table, build as many different arrangements as you can that close around one point with no gaps. Every piece must have a corner at the point. A new arrangement must use different numbers of triangles or rhombi. Sketch each successful arrangement and write its two counts.





Triangles: ____ Rhombi: ____



Triangles: ____ Rhombi: ____



Triangles: ____ Rhombi: ____



Triangles: ____ Rhombi: ____



Triangles: ____ Rhombi: ____



Triangles: ____ Rhombi: ____

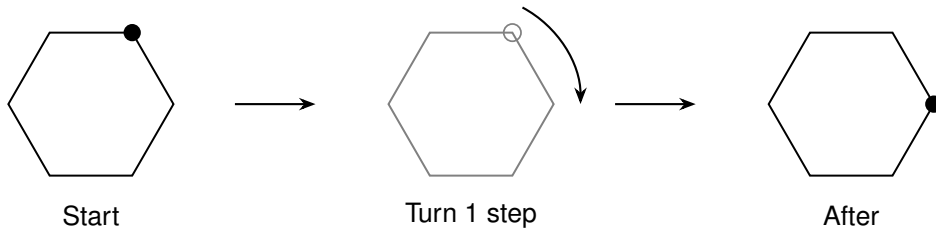


Triangles: ____ Rhombi: ____



Triangles: ____ Rhombi: ____

A 1-step turn moves each corner clockwise to the next corner. The blocks and their inside edges turn together.



Problem 3: On the table, fill a regular hexagon with these blocks. Its outer sides are each one block side. Which turns make your design match itself, including every edge inside? Make designs with different answers. Sketch each design and circle every turn that works.

